

[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

May-26-0207

**MCA-6205 (Web Technologies (Node.js and MongoDB)
MCA 2nd (NEP)**

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all selecting one question each from Unit I, II, III and IV. Question 9 is compulsory.

UNIT - I

1. Explain the role of CSS in web development. Discuss CSS properties and styling pages with examples. (12)
2. What is JavaScript? Write a program to validate a user registration form (name, email, and password) using JavaScript. (12)

UNIT - II

3. Describe the working of Node.js with a neat diagram of its process model. How is it different from traditional client-server architecture? (12)
4. What is Node Package Manager (NPM)? Explain steps for installing, updating, and removing packages both locally and globally. (12)

UNIT - III

5. Write a program in Node.js to:
 - (a) Open a file
 - (b) Read its contents
 - (c) Append new data into it(3×4=12)

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6. Explain debugging in Node.js. How does the Node.js debugger work? Write an example demonstrating breakpoints and step execution. (12)

UNIT - IV

7. Discuss the steps for connecting Node.js with MongoDB using Mongoose. Write code to create a new database connection. (12)
8. Explain CRUD operations in detail. Write Node.js code snippets for:
 - (a) Inserting a document
 - (b) Updating a document
 - (c) Deleting a document(3×4=12)

UNIT - V (Compulsory)

9. Short Answer Questions.
 - (a) Define HTML5 Canvas with an example use case.
 - (b) Write one difference between synchronous and asynchronous file operations.
 - (c) What is the function of require() in Node.js?
 - (d) Give one advantage and one disadvantage of Node.js.
 - (e) Mention any two uses of EventEmitter in Node.js.
 - (f) Write the MongoDB command for displaying all documents in a collection. (6×2=12)